

1.RESEARCH QUESTION

Does the frequency of weekly aerobic exercise significantly differ across adolescent grade levels (9th to 12th grade)?

2. Variable Definitions

Independent Variable (X): Grade

Dependent Variable (Y): Aerobic_Exercise

3.Statistical Method

One-Way ANOVA
Tukey HSD Post-Hoc Test

Null Hypothesis (H0): The population mean days of weekly aerobic exercise are equal across all high school grade levels.
Alternative Hypothesis (H1): At least one grade level has a different population mean days of weekly aerobic exercise from the others.

4. Statistic result

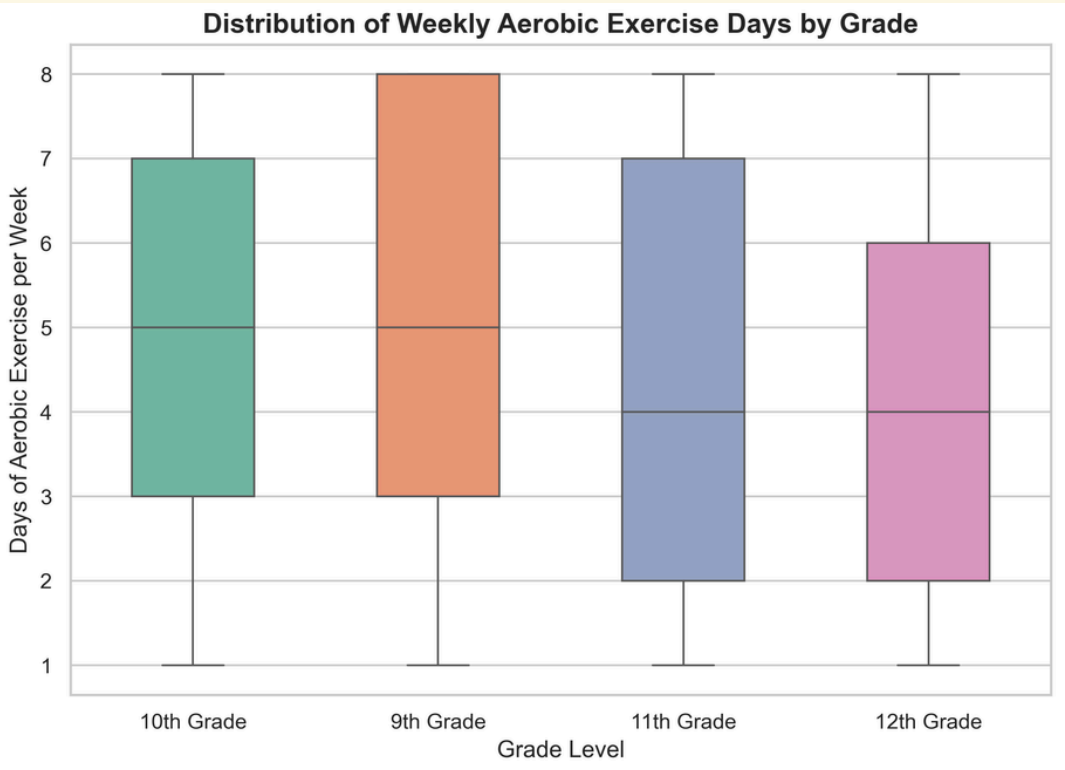
One-Way ANOVA :
ANOVA Result: p-value: 2.086895e-52 (p < 0.001)

===== ANOVA 檢定結果 =====				
	sum_sq	df	F	PR(>F)
C(Grade)	1514.134816	3.0	81.809265	2.086895e-52
Residual	76728.451311	12437.0	NaN	NaN

Tukey HSD Post-Hoc Test:

===== Tukey HSD 事後比較結果 =====						
Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
10th Grade	11th Grade	-0.3369	0.0	-0.4992	-0.1747	True
10th Grade	12th Grade	-0.6651	0.0	-0.8264	-0.5039	True
10th Grade	9th Grade	0.2585	0.0003	0.0945	0.4225	True
11th Grade	12th Grade	-0.3282	0.0	-0.488	-0.1685	True
11th Grade	9th Grade	0.5954	0.0	0.4329	0.758	True
12th Grade	9th Grade	0.9236	0.0	0.7621	1.0852	True

5. figure



6. conclusion

A One-Way ANOVA and Tukey’ s HSD test on the YRBS dataset evaluated the impact of high school grade levels on weekly aerobic exercise. The results reveal highly significant differences across grades, strongly rejecting the null hypothesis.